

Geography Playbook

Maps & Globe · Landforms · Water Bodies · Agriculture · Minerals · North & South America

ICSE Class 6 · Full Book · Final Exam Ready

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Chapter names and order can differ slightly between schools and publishers. Follow your school's exam syllabus.

How to use this playbook

Unit	Chapters
Representation of features	1. Globe, Latitudes and Longitudes · 2. Maps
The Earth's surface	3. Landforms: Mountains, Plateaus, Plains · 4. Water Bodies
Resources	5. Agriculture · 6. Minerals
Continents	7. North America · 8. South America

⭐ **Exam Tip: How to score full marks in Geography**

1. **Map work carries easy marks.** Practise marking rivers, mountains, lakes and countries on outline maps every week.
2. **Use an atlas** for countries and capitals. The ICSE asks for these from the atlas.
3. "Give reasons" answers: **cause → effect** (because ..., therefore ...).
4. Learn the **"est" facts** (longest, highest, largest). They are favourite one-mark questions.

1. The Globe, Latitudes and Longitudes

- A **globe** is a **true model** of the Earth. It shows the correct shape, size and direction, but it isn't handy to carry around.
- The Earth's **axis** is an imaginary line through the **North and South Poles**. It is **tilted at 23½°** from the vertical.
- The Earth is not a perfect sphere. It is slightly flattened at the poles: a **geoid**.

Latitudes (parallels)

- Imaginary **horizontal circles** drawn **parallel to the Equator**.
- The **Equator (0°)** divides the Earth into the **Northern and Southern Hemispheres**. It is the longest latitude.
- Latitudes get **smaller towards the poles**. The poles (90° N and 90° S) are just **points**.

Important latitude	Degree
Equator	0°
Tropic of Cancer	23½° N
Tropic of Capricorn	23½° S
Arctic Circle	66½° N
Antarctic Circle	66½° S
North Pole / South Pole	90° N / 90° S

- **Heat zones: Torrid** (hottest, between the two tropics), **Temperate** (between the tropics and the polar circles), **Frigid** (coldest, beyond the polar circles)

Longitudes (meridians)

- Imaginary **half-circles** that join the **North and South Poles**. All longitudes are **of equal length**.
- The **Prime Meridian (0°)** passes through **Greenwich, near London**. It divides the Earth into the **Eastern and Western Hemispheres**.
- There are **360 longitudes**: 180° E and 180° W. The 180° line is roughly the **International Date Line**.
- **Time:** the Earth turns 360° in 24 hours, which is **15° in 1 hour**, or **1° in 4 minutes**.
- **Indian Standard Time (IST)** is based on **82½° E** (near Mirzapur, U.P.). IST = GMT **+ 5 hours 30 minutes**.

Grid

- The network of latitudes and longitudes is called the **grid**. It helps to find the **exact location** of any place.

🧠 Memory Trick

LATitude = LADDER (horizontal rungs, going up and down). **LONGitude = LONG lines from pole to pole. "Cancer is up north, Capricorn is down south"** (C-up / C-down: "Cancer Climbs, Capricorn Comes down").

📍 Mumbai Connection

Mumbai is at about **19° N, 72.8° E**. It lies **south of the Tropic of Cancer**, so it is in the **Torrid Zone** and stays warm all year!

🎲 Play It in Class: Orange Globe

Give each team an orange. They draw the Equator, the tropics and the polar circles with a marker, then draw the Prime Meridian from "pole" to "pole". Cut it open: do the latitude rings get smaller near the top? Yes!

✏️ Quick Check

1. What is the tilt of the Earth's axis?
2. Name the latitude at (a) 23½° N (b) 66½° S.
3. Why is IST 5 hours 30 minutes ahead of GMT? (Hint: 82½ × 4 minutes)
4. What is a grid?

2. Maps

- A **map** is a drawing of the Earth's surface, or a part of it, on a **flat surface**, drawn to **scale**.
- **Map vs globe:** a map is easy to carry and can show small areas in detail, but it **distorts** shapes and sizes a little.
- **Cartography** is the art of making maps. A **cartographer** is a map-maker. An **atlas** is a book of maps.

Types of maps

Type	Shows	Example
Physical (relief)	Natural features: mountains, rivers, plains	A physical map of India
Political	Boundaries of countries and states, cities	A map of Indian states
Thematic	One particular theme	Maps of rainfall, crops, population

Components of a map

1. **Title:** what the map is about
2. **Direction:** the north arrow (N). The **cardinal directions** are N, S, E and W, and the **intermediate directions** are NE, NW, SE and SW.
3. **Scale:** the ratio between a distance on the map and on the ground
   - Statement scale: "1 cm = 10 km"
   - Representative fraction (RF): 1 : 1,00,000
   - Linear (graphic) scale: a line divided into parts
4. **Legend (key):** explains the symbols and colours used
5. **Conventional signs and symbols:** the same everywhere (a railway line, a temple, a post office)

Colours on a physical map

Colour	Shows
Blue	Water bodies
Green	Lowlands, plains
Yellow and brown	Highlands, plateaus, mountains
White	Snow-covered peaks

- **Sketch:** drawn **without scale**, from memory. **Plan:** a **large-scale** drawing of a small area (a house or a school).

> 🧠 **Memory Trick**
>
> **Map components: "TDSLC = Ten Dancing Snakes Love Chocolate"**: **T**itle, **D**irection, **S**cale, **L**egend, **C**onventional signs. **Directions: "Never Eat Soggy Waffles"** (clockwise from N: North, East, South, West).

🎲 Play It in Class: Draw Your Route Map

Each student draws a map from home to school with a title, a north arrow, a rough scale and a legend (🏫 school, 🚉 station, 🛕 temple, 🌳 park). Partners then try to "follow" each other's maps!

✏️ Quick Check

1. Define a map. How is it different from a globe?
2. Name the five components of a map.
3. What does blue show on a physical map? What does brown show?
4. What is the difference between a plan and a sketch?

3. Landforms: Mountains, Plateaus and Plains

- **Landforms** are the natural features of the Earth's surface. The three main ones are **mountains, plateaus and plains**.

Mountains

- A **mountain** is a landform that rises **steeply** from the surrounding land, with a **peak (summit)**. It is generally higher than about **600 m**.
- A **range** is a line of mountains. A **valley** is the low land between mountains.

Type	How it formed	Examples
Fold mountains	Rock layers were **pushed and folded** by forces inside the Earth. They are young and high.	**Himalayas**, Alps, **Andes**, **Rockies**
Block mountains	Land **lifted up** between cracks (faults). The sunken part forms a **rift valley**.	Vosges and Black Forest (Europe), **Satpura** (India)
Volcanic mountains	Lava piled up around an opening	Mt Fuji (Japan), Mt **Kilimanjaro** (Africa), Vesuvius (Italy)
Residual mountains	Old mountains **worn down** by erosion	**Aravallis**, Nilgiris (India)

- **Rift valley:** a long, narrow valley formed when land sinks between two faults: the **Great Rift Valley** (Africa), the **Narmada valley** (India).
- **Life in the mountains:** a cold climate, thin air, little farming (**terrace farming** on slopes), tourism, few roads.

Plateaus

- A **plateau** is a **raised, flat-topped** land with steep sides. It is called a "**table-land**".

Type	Example
Intermontane (surrounded by mountains)	**Tibetan Plateau**, the **highest** plateau ("Roof of the World")
Piedmont (at the foot of mountains)	**Patagonia** (South America), Malwa (India)
Continental (rises from the plains or sea, a large area)	**Deccan Plateau** (India)

- Plateaus are **rich in minerals** (Chota Nagpur is the "mineral heartland of India") and have **waterfalls** (Jog Falls, Hundru Falls).

Plains

- **Plains** are **flat, low-lying** lands, usually less than 200 m high.
- **Depositional (alluvial) plains** are formed by rivers depositing silt: the **Indo-Gangetic Plain**, the Mississippi Plain.
- **Plains are the most populated areas**: fertile soil, easy farming, easy to build roads, railways and cities.

🧠 Memory Trick

Mountain types: "FBVR = Five Big Volcanic Rocks": **F**old, **B**lock, **V**olcanic, **R**esidual. **Mountain = an ice-cream cone (peak). Plateau = a table (flat top). Plain = the floor (flat and low).**

📍 Mumbai Connection

Mumbai sits on the **western edge of the Deccan Plateau**, which is made of **basalt** rock from ancient lava flows. The **Western Ghats (Sahyadri)**, near Lonavala, Matheran and Mahabaleshwar, are the steep edge of this plateau. The **Konkan coastal plain** lies between the Ghats and the sea.

🎲 Play It in Class: Clay Landforms

Teams model a fold mountain (push layers of coloured dough together from both sides), a block mountain (lift up the middle piece), a plateau (a flat-topped block) and a plain. They label each one with an example.

✏️ Quick Check

1. Name the four types of mountains, with one example each.
2. Why is the Tibetan Plateau called the "Roof of the World"?
3. Why are plains densely populated?
4. What is a rift valley? Give an example.

4. Water Bodies

- About **71%** of the Earth is covered by water. That's why the Earth is called the **Blue Planet**.

Oceans

Ocean	Facts
Pacific	The **largest and deepest**. The **Mariana Trench** (about 11,000 m) is the deepest point on Earth.
Atlantic	The second largest, **S-shaped**, with the busiest shipping routes
Indian	The only ocean **named after a country**
Southern (Antarctic)	Surrounds Antarctica
Arctic	The **smallest and shallowest**, frozen for most of the year

Seas, bays and gulfs

- A **sea** is a part of an ocean, partly enclosed by land.
- **Marginal sea:** on the edge of an ocean (the **Arabian Sea**, South China Sea).
- **Inland or enclosed sea:** almost surrounded by land (the **Mediterranean Sea**, Black Sea).
- **Bay:** a wide curved inlet (the **Bay of Bengal**). **Gulf:** a deep, narrow inlet (the Gulf of Mannar, Gulf of Mexico).
- **Strait:** a narrow water passage joining two seas (the **Palk Strait**). **Isthmus:** a narrow strip of land joining two land masses (the **Isthmus of Panama**).

Lakes

Record	Lake
Largest lake (saltwater)	**Caspian Sea**
Largest freshwater lake	**Lake Superior** (North America)
Deepest lake	**Lake Baikal** (Russia)
Highest navigable lake	**Lake Titicaca** (South America)
Largest lagoon in India	**Chilika** (Odisha)

Rivers

- **Source:** where the river begins. **Mouth:** where it meets the sea.
- **Tributary:** a smaller river that **joins** the main river. **Distributary:** a branch that **leaves** the main river near the mouth.
- **Delta:** a triangular area of deposited silt at the mouth (the **Sundarbans delta**, the Ganga–Brahmaputra). **Estuary:** a wide, deep mouth with no delta (the **Narmada and Tapi**).
- **Meander:** a loop in a river. An **oxbow lake** is a cut-off meander.
- **Longest river:** the **Nile** (Africa). **Largest by volume:** the **Amazon** (South America).

🧠 Memory Trick

The 5 oceans by size: "Please Adjust Indian Shoes Accordingly": **P**acific, **A**tlantic, **I**ndian, **S**outhern, **A**rctic. **Tributary = joins in ("tribute" = gives water). Distributary = distributes (leaves).**

📍 Mumbai Connection

Mumbai is on the **Arabian Sea** (a marginal sea of the Indian Ocean). It was once **seven islands** that were joined together! The **Mithi River** flows through the city and enters the sea at **Mahim Creek**. **Powai, Vihar and Tulsi** are lakes inside Mumbai.

🎲 Play It in Class: Water Body Bingo

Make bingo cards with words such as delta, strait, bay, isthmus, tributary, Pacific, Caspian, Baikal. The teacher reads out definitions ("the deepest lake in the world"). Students mark the answer. The first full line wins!

✏️ Quick Check

1. Name the largest and the smallest ocean.
2. Differentiate between a tributary and a distributary.
3. What is the difference between a strait and an isthmus?
4. Name the deepest lake in the world.

5. Agriculture

- **Agriculture** is growing **crops** and rearing **animals**. More than **half of India's** workers depend on it.
- **Factors affecting agriculture:** climate (temperature, rainfall), soil, relief (flat land), water (irrigation), and human factors (labour, machines, markets).

Types of agriculture

Type	Features	Example
Subsistence	Grown **for the family's own use**, on small farms with simple tools	Small Indian villages
Commercial	Grown **for sale**, on large farms with machines	Wheat farms in the USA and Canada
Plantation	**A single cash crop** on a big estate	Tea in Assam, coffee in Karnataka, rubber in Kerala
Shifting (slash and burn)	Clear the forest, farm, then **move on** when the soil loses its fertility	**Jhum** (North-East India), the Amazon basin
Mixed farming	Crops + animal rearing together	Europe, Punjab
Intensive	Small land, a lot of labour, high output per acre	Rice in India, China
Extensive	Large land, machines, less labour	The Prairies (North America)

Crop seasons of India

Season	Sown	Harvested	Crops
Kharif	June–July (monsoon)	Sept–Oct	**Rice**, cotton, maize, jowar, bajra, soybean
Rabi	Oct–Dec (winter)	March–April	**Wheat**, mustard, gram, barley
Zaid	March–June (summer)	June	Watermelon, cucumber, vegetables

Major crops

Crop	Conditions	Major areas
Rice	High temperature (above 25 °C), **heavy rain (100–200 cm)**, clayey soil, standing water	West Bengal, U.P., Punjab, **Konkan**; China, Bangladesh
Wheat	Cool, moist growing season, **moderate rain (50–75 cm)**, a sunny ripening period, loamy soil	Punjab, Haryana, U.P., M.P.; the USA and Canada Prairies
Cotton	Warm, frost-free, moderate rain, **black soil (regur)**	**Maharashtra (Vidarbha)**, Gujarat; the USA
Sugarcane	Hot and humid, plenty of water	U.P., **Maharashtra (Kolhapur, Sangli)**; Brazil
Tea	Warm, heavy rain, **well-drained hill slopes**	Assam, Darjeeling; Sri Lanka, China
Coffee	Warm, shady hill slopes	Karnataka; **Brazil** (the largest producer)

- **Food crops:** rice, wheat, millets, pulses. **Cash crops:** cotton, jute, sugarcane, tea, coffee.
- **Green Revolution (1960s):** high-yielding seeds, fertilisers and irrigation greatly increased **wheat and rice** production in India.

🧠 **Memory Trick**

"KR-Z = Kharif Rain, Rabi wintER, Zaid summer" "Rice loves Rain, Wheat loves Winter."

📍 **Mumbai Connection**

The **Alphonso (Hapus) mango** from Ratnagiri and Devgad and the **rice fields of the Konkan** grow near Mumbai. Mumbai's **Vashi APMC market** (in Navi Mumbai) is one of Asia's biggest wholesale markets for fruits and vegetables!

🎲 **Play It in Class: Grow-a-Farm Cards**

Teams get a "climate card" (hot + heavy rain + clay soil, or cool + moderate rain + loamy soil…). They must choose the best crop and season, then place it on an India map. Correct crop + correct region = 2 points.

✏️ **Quick Check**

1. Distinguish between subsistence and commercial agriculture.
2. Name two kharif and two rabi crops.
3. Write the conditions needed to grow rice.
4. What is shifting cultivation? What is it called in North-East India?

6. Minerals

- A **mineral** is a **natural substance** with a definite chemical composition, found in the Earth's crust. **Ores** are rocks from which minerals are taken out profitably.

Classification

Metallic		Non-metallic	Fuel (energy) minerals
Ferrous (contain iron)	**Non-ferrous** (no iron)		
Iron ore, manganese	Copper, bauxite (aluminium), gold, silver	Mica, limestone, gypsum, salt	**Coal, petroleum, natural gas** (fossil fuels)

Important minerals and their uses

Mineral	Uses	Where in India
Iron ore	Steel making (buildings, machines)	Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa
Bauxite	**Aluminium** (aircraft, utensils, foil)	Odisha, Gujarat, Jharkhand
Copper	Electric wires (it is a good conductor)	Rajasthan (Khetri), Jharkhand
Mica	Electrical insulation	Jharkhand, Andhra Pradesh
Limestone	Cement	M.P., Rajasthan
Coal	Fuel for thermal power and the iron and steel industry	Jharkhand (Jharia), Odisha, Chhattisgarh
Petroleum	Fuel (petrol, diesel), plastics	**Mumbai High** (offshore), Assam (Digboi), Gujarat

- **Mining methods:** open-cast (surface), shaft (deep underground), drilling (oil and gas)
- **Conservation:** minerals are **non-renewable**, so reduce, reuse and recycle (recycle aluminium cans and scrap iron) and use substitutes.

🧠 **Memory Trick**

Ferrous = Fe = iron. "If it has **Fe**, it's **Fe**rrous!" **Bauxite → Aluminium: "B for Bauxite, B for Bottles and Bikes made of aluminium."**

📍 **Mumbai Connection**

Mumbai High is an offshore oil field in the Arabian Sea, about **160 km off the Mumbai coast**, discovered in **1974**. It is one of India's most important oil fields!

🎲 Play It in Class: Mineral Hunt at Home

Students list 10 things at home and find out which mineral each one comes from: steel spoon (iron), foil (bauxite), electric wire (copper), cement wall (limestone), LPG (petroleum/natural gas), salt (rock salt or seawater). Share and make a class chart.

✏️ Quick Check

1. What is the difference between ferrous and non-ferrous minerals?
2. Which mineral is used to make aluminium?
3. Name one place in India where petroleum is found.
4. Why should we conserve minerals?

7. North America

Location and size

- The **3rd largest continent**. It lies entirely in the **Northern and Western Hemispheres**.
- **Boundaries:** the **Arctic Ocean** (north), the **Atlantic Ocean** (east), the **Pacific Ocean** (west), and the **Gulf of Mexico and Caribbean Sea** (south-east).
- It is joined to South America by the **Isthmus of Panama**. The **Panama Canal** connects the Atlantic and Pacific Oceans.
- **Greenland**, the **largest island** in the world, is a part of North America.

Political divisions (learn these from the atlas)

- **Canada** (the 2nd largest country in the world), the **USA**, **Mexico**, the Central American countries and the Caribbean islands (Cuba, Jamaica...)

Physical divisions

Region	Features
Western Cordillera	Young fold mountains: the **Rockies**. The highest peak is **Denali (Mt McKinley)**. Plateaus such as Colorado, with the **Grand Canyon**. **Death Valley** is the lowest and hottest place.
Great Central Plains	Flat, fertile plains: the **Prairies** (grasslands, the "**bread basket of the world**", famous for **wheat**)
Eastern Highlands	Old, worn-down mountains: the **Appalachians**
Canadian (Laurentian) Shield	Ancient hard rocks around **Hudson Bay**, with many lakes and rich minerals

Rivers and lakes

- **Mississippi–Missouri:** the longest river system of North America. It flows into the **Gulf of Mexico**.
- **St. Lawrence** (drains the Great Lakes into the Atlantic), **Mackenzie** (flows into the Arctic), **Colorado** (formed the Grand Canyon), **Yukon**, **Rio Grande** (the USA–Mexico border)
- **The Great Lakes: Superior** (the largest freshwater lake in the world), **Michigan**, **Huron**, **Erie**, **Ontario**. **Niagara Falls** lies between Lake Erie and Lake Ontario.

Climate and vegetation

- The far north is **tundra** (very cold: mosses, lichens, the **Inuit** people). Canada has **coniferous forests** (softwood trees). The south is warmer: grasslands and deserts.

Lumbering in Canada (case study)

- Canada has huge **coniferous forests** of **pine, fir and spruce** with **soft wood**, which is good for **paper, pulp and furniture**.
- **Why lumbering is easy:** pure stands of trees (one kind together), frozen ground in winter for moving logs, rivers to **float logs** down to the mills in spring, cheap hydroelectric power and

machines.

- **Conservation:** afforestation (planting new trees) and controlled cutting.

🧠 **Memory Trick**

The Great Lakes: "HOMES": **H**uron, **O**ntario, **M**ichigan, **E**rie, **S**uperior. **From west to east: "Rockies → Prairies → Appalachians" = "Young → Flat → Old"**.

🎲 **Play It in Class: Atlas Race**

Call out a feature ("Lake Huron!", "Rockies!", "Panama Canal!"). Pairs race to find it in their atlas and point to it. The first correct pair wins a point. Then mark 10 features on an outline map of North America.

✏️ **Quick Check**

1. Name the oceans that surround North America.
2. Name the five Great Lakes. Which is the largest?
3. Why are the Prairies called the "bread basket of the world"?
4. Give three reasons why lumbering is developed in Canada.

8. South America

Location and size

- The **4th largest continent**. The **Equator** passes through its northern part (**Ecuador, Brazil**), and the **Tropic of Capricorn** crosses it further south. So it lies in both hemispheres, mostly in the **Southern** one.
- **Boundaries:** the **Caribbean Sea** (north), the **Atlantic Ocean** (east), the **Pacific Ocean** (west). The southern tip is **Cape Horn**, by the **Drake Passage**.

Political divisions (learn these from the atlas)

- **Brazil** (the largest country), Argentina, Peru, Colombia, Chile, Venezuela, Bolivia, Ecuador, Paraguay, Uruguay, Guyana, Suriname

Physical divisions

Region	Features
The Andes (west)	The **longest mountain range in the world** (young fold mountains). The highest peak is **Mt Aconcagua** (Argentina). **Lake Titicaca** is on the Peru–Bolivia border.
Central plains	**Amazon basin** (hot, wet rainforest), the **Llanos** (Orinoco grasslands), the **Pampas** (Argentina, temperate grasslands for wheat and cattle)
Eastern Highlands	**Brazilian Highlands** and **Guiana Highlands** (where **Angel Falls**, the **highest waterfall in the world**, is in Venezuela)
Patagonia	A dry plateau in southern Argentina
Atacama Desert	In northern Chile, **one of the driest places on Earth**

Rivers

- **Amazon:** the **largest river in the world by volume**. It flows west to east into the Atlantic and has more than 1,000 tributaries.
- **Orinoco** (Venezuela), **Paraná–Paraguay** (flows south into the **Rio de la Plata**)

Life in the Amazon basin (case study)

- **Climate: hot and wet all year** (equatorial), with rain almost every afternoon.
- **Vegetation: dense tropical rainforests** called **selvas**: tall evergreen trees, the "**lungs of the Earth**".
- **Animals:** jaguar, anaconda, **piranha**, sloth, toucan, macaw, tree frogs.
- **People:** tribes who hunt, fish, gather and do **shifting cultivation** (they grow cassava). Their houses have sloping roofs, sometimes on stilts. They travel by **boat** on the rivers.
- **Problems: deforestation** for farming, cattle ranching, roads and mining. It threatens wildlife, tribal life and the world's climate.

Grasslands of South America

Name	Country
Llanos	Venezuela and Colombia
Campos	Brazil
Pampas	Argentina

🧠 Memory Trick

"Amazon = Amazing volume. Andes = Always long." Grasslands: "Lazy Cows Pamper themselves": **L**lanos, **C**ampos, **P**ampas (from north to south).

📍 Mumbai Connection

Brazil is the world's **largest coffee producer**, so there's a good chance some of the coffee in Mumbai's cafés has travelled from Brazil! Also, the Amazon rainforest and Mumbai's **Sanjay Gandhi National Park** are both green "lungs", just of very different sizes.

🎲 Play It in Class: Amazon Explorer Diary

Each student writes a 5-line "explorer diary" about a day in the Amazon: the weather, the animals they saw, how the people live, and one danger to the forest. Read the best entries aloud.

✏️ Quick Check

1. Name the longest mountain range and the highest waterfall in South America.
2. Why is the Amazon basin hot and wet all year?
3. Name the grasslands of Argentina and Venezuela.
4. Give two reasons why the Amazon rainforest must be protected.

Full-Book Revision Test

Section A (1 mark each)

1. Which latitude is at 0°?
2. What does the colour green show on a physical map?
3. Name a residual mountain in India.
4. Which is the largest freshwater lake in the world?
5. Name one rabi crop.
6. Where is Mumbai High?

Section B (2–3 marks each)

7. Differentiate between latitudes and longitudes (three points).
8. Name the three types of scale, with an example of each.
9. Explain how fold mountains are formed. Give two examples.
10. Differentiate between a delta and an estuary.
11. Write the conditions needed for growing wheat.
12. Name two ferrous and two non-ferrous minerals.

Section C (4 marks each)

13. Describe the physical divisions of North America.
14. Explain life in the Amazon basin (climate, vegetation, people, problems).
15. On an outline map of South America, mark: the Andes, the Amazon river, Lake Titicaca, the Pampas, the Atacama Desert, Brazil.

Answer Key

Ch 1: (1) 23½° (2) (a) The Tropic of Cancer (b) The Antarctic Circle (3) IST is based on 82½° E, and 82½ × 4 minutes = 330 minutes = 5 hours 30 minutes. (4) The network of latitudes and longitudes used to find exact locations.

Ch 2: (1) A drawing of the Earth's surface, or a part of it, on a flat surface, drawn to scale. It is easy to carry and shows details, but it is less accurate than a globe. (2) Title, direction, scale, legend, conventional signs (3) Blue shows water. Brown shows highlands and mountains. (4) A plan is drawn to a large scale for a small area. A sketch is drawn without a scale, from memory.

Ch 3: (1) Fold (Himalayas), block (Satpura or Vosges), volcanic (Kilimanjaro), residual (Aravallis) (2) It is the highest plateau in the world. (3) They have fertile soil, are easy to farm, and are easy to build transport and cities on. (4) A long, narrow valley formed when land sinks between two faults: the Great Rift Valley or the Narmada valley.

Ch 4: (1) Largest: the Pacific. Smallest: the Arctic. (2) A tributary joins the main river. A distributary branches out of it near the mouth. (3) A strait is a narrow passage of water joining two seas. An isthmus is a narrow strip of land joining two land masses. (4) Lake Baikal

Ch 5: (1) Subsistence agriculture is for the family's own use, on small farms. Commercial agriculture is for sale, on large mechanised farms. (2) Kharif: rice, cotton. Rabi: wheat, mustard. (3) Above 25 °C, 100–200 cm of rain, clayey soil, standing water (4) Clearing and burning forest to farm, then moving on. It is called **jhum** in North-East India.

Ch 6: (1) Ferrous minerals contain iron. Non-ferrous minerals don't. (2) Bauxite (3) Mumbai High, Digboi (Assam) or Gujarat (4) They are non-renewable and will run out.

Ch 7: (1) The Arctic, Atlantic and Pacific (2) Huron, Ontario, Michigan, Erie, Superior. Superior is the largest. (3) Their huge fertile plains produce large amounts of wheat. (4) Pure stands of softwood trees, frozen ground in winter, rivers to float logs, cheap hydroelectric power and machines (any three)

Ch 8: (1) The Andes and Angel Falls (2) It lies on or near the Equator, so it gets direct sunlight and convectional rain every day. (3) Argentina: the Pampas. Venezuela: the Llanos. (4) It produces oxygen and controls the climate, and it is home to millions of species and tribal people.

Revision test: (1) The Equator (2) Lowlands and plains (3) The Aravallis or Nilgiris (4) Lake Superior (5) Wheat, mustard or gram (6) In the Arabian Sea, off the Mumbai coast (7) Latitudes are parallel circles of decreasing length from east to west. Longitudes are half-circles of equal length running from pole to pole. Latitudes show heat zones, and longitudes show time. (8) Statement (1 cm = 5 km), RF (1 : 5,00,000), linear (a divided line) (9) Rock layers are compressed and folded by forces inside the Earth. Examples: the Himalayas, the Alps. (10) A delta is a triangular deposit of silt at the mouth (the Ganga). An estuary is a wide, deep mouth without deposits (the Narmada). (11) A cool, moist growing season, 50–75 cm of rain, a sunny ripening period, loamy soil (12) Ferrous: iron ore, manganese. Non-ferrous: copper, bauxite. (13) The Western Cordillera (Rockies), the Great Central Plains (Prairies), the Eastern Highlands (Appalachians), the Canadian Shield (14) Hot, wet equatorial climate. Dense selvas forests. Tribes hunting, fishing and doing shifting cultivation, travelling by boat. Problems: deforestation. (15) Check against the atlas.